

Math 166

Name: _____

New series from old

Section Number: _____

1. Find the *radius of convergence* and *interval of convergence* of the power series

$$\sum_{n=0}^{\infty} (-1)^n \frac{n}{4^n} (x+1)^{2n}.$$

2. Find the radius (and interval) of convergence for each of the following power series:

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ and } \sum_{n=0}^{\infty} n! x^n.$$

3. A series of the form $\sum_{n=0}^{\infty} a_n x^n$ is known to converge at $x = 4$. For each of the following values, determine whether the series must converge, must diverge, or it is not possible to determine. Justify your answer.

(a) $x = -3$.

(b) $x = -4$.

(c) $x = 5$.

4. Find the power series for $\arctan x$. (Hint: what is the derivative of $\arctan x$? Express this in the form $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$.)

5. Use the answer to your previous problem to find a series that converges to π . (Hint: what is $\arctan 1$?)